

Batch Normalisation: Batch normalisation is a technique that normalises or brings all the layer inputs to the same reduced unit. This helps to speed up training and apply different learning rates. Batch normalisation is a highly effective tool for convolutional and feedforward neural networks but lacks the potential for recurrent or feedback loops systems.

Highway Layer: A highway layer is a neural network program that employs a gateway procedure to control the transfer of information through layers, like a filter. Stacking multiple layers can allow the system to learn and train very large and deep networks.

Momentum: Momentum is an extendible portion to the gradient descent algorithm that can either reduce or increase the number of items the inputs are updated after they pass through the nodes.

Epoch: Most neural network softwares use epoch to represent the number of iterations that the user wants the network to work for. Increasing epochs will increase the computational needs but can yield better accuracies as the system takes more time to learn the data.

Relational Coefficient: It's a way of comparing the expected and actual output, often represented by the letter R on neural network graphs- represented between 0 and 1. A larger value of R indicates that the model works well for the data but be wary of models that yield a complete value of 1 since this indicates that the model is using too much training data and not enough testing data.

ROC Curves: The reactor operating curve is used to compare multiple neural network schemes and see where they best fit the expected outcome curve and at what iteration. For classification cases, the output node is usually modified until a good fit is found.

Getting through with neural networks

There's a lot of data out there for people to use and to be worked on. Just searching for public research datasets online will open up vast repositories that can be used and sampled to the actual results. Many of these repositories might be owned separately by academic institutions or public sources.

One such place that offers data for free happens to be the UCI repository with some real-life dataset examples that can be applied to neural networks.

1. To work on a neural network, it's important first to master the syntax of the computing environment where the network will be built. While the steps will match for all environments, the keyword and phrases will be different.